

Smart Edge for Cloud-Ready Data

Connecting a device is no longer enough to call it 'smart'. The real challenge lies in real-time data refining: operating on milliwatts while ensuring total privacy at the edge

Des données orientées Cloud directement depuis les systèmes embarqués Dati orientati al Cloud direttamente dai dispositivi integrati

Below are some critical challenges — and opportunities — where embedded systems design directly impacts the effectiveness of real-time data engineering at the edge

DATA COLLECTION (SOURCE)

Capturing raw signals at the point of origin

DATA INGESTION (PIPELINE)

Efficiently moving bits into local memory

DATA PROCESSING (REFINING)

Transforming noisy waveforms into structured, meaningful features

DATA QUALITY (VALIDATION)

Ensuring integrity before any data leaves the device

DATA STORAGE (PERSISTENCE)

Local buffering and temporary storage for fault-tolerant streaming

DATA ANALYTICS (INTELLIGENCE)

Lightweight computation at the milliwatt level, near the sensor

DATA TRANSFORMATION (SYNTHESIS)

Compressing and enriching data to high-value payloads

DATA INTEGRATION (TRANSPORT)

Sending only refined, relevant data to the cloud

DATA GOVERNANCE (LIFECYCLE)

Managing versions, metadata, and signal health on the edge

Cloud Integration Map

EDGE FUNCTION	AWS STACK	GCP STACK
Ingestion	IoT Core / Kinesis	IoT Core / Pub-Sub
Processing	Lambda / Glue	Cloud Functions / Dataflow
Analytics	Redshift/Timestream / S3	BigQuery / Cloud Storage
Governance	IoT Device Management	Fleet Management

Refining the Cloud pipeline at the Edge: Intelligence at the point of origin

Sensors & Real-Time Data Processing

HIGH-SPEED DATA STREAMING

Low-latency data pipelines targeting deterministic sub-millisecond latency — an ideal design goal for real-time, cloud-native processing at the edge

HANDLING RAW BINARY SIGNALS

On-device data validation and filtering to remove noise and outliers before cloud ingestion

EDGE COMPUTING ARCHITECTURE

Edge-native architectures enabling local synthesis and significant bandwidth reduction before data reaches the cloud

Data Efficiency & Payload Optimization

SMART DOWNSAMPLING

Reducing data volume while preserving informational entropy

LOCAL AGGREGATION

Transforming continuous raw signals into discrete, high-level events

Hardware-Level Security & Data Privacy

DATA ANONYMIZATION

On-chip hashing and encryption to enforce privacy by design

ZERO RAW-DATA CLOUD EXPOSURE

Raw signals remain on-device, with only refined and contextualized insights transmitted upstream

TinyML: On-Device Data Intelligence

FEATURE ENGINEERING

Extracting meaningful features directly on constrained devices

ANOMALY DETECTION

Local pattern recognition to detect abnormal behavior in real time

QUANTIZATION

Reducing model size and compute requirements while preserving accuracy

Towards Intelligent Edge

Improving cloud data processes starts with better data at the source

Which of these challenges is most critical for your project?

Quel point est critique pour votre projet ?

Quale di questi punti è critico per il tuo progetto ?